CHALLENGE

02

Voice That Sets People Free

Don't build an assistant that answers. Build one that anticipates.

POWERED BY

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Voice AI partners

Hacktivate 2026 · 24-hour Hardware + AI Hackathon · Theme: Predictive Systems
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Goals and Motivation

Meet Shankar

Shankar is 74 and lives with his son's family in Basavanagudi. Since a fall last year he walks with a walker, and his eyesight has faded enough that he can no longer read the labels on his tablet strips. On weekdays his son and daughter-in-law leave for work at 8:30, and the grandchildren leave for school. From then until evening, Shankar is at home alone.

His day has a shape everyone in the family knows. Tablets after breakfast and after lunch. A cup of tea at 4. A slow walk from his chair to the bathroom every couple of hours, along a corridor that is dim in the afternoon. A nap after lunch that sometimes runs too long. His daughter-in-law calls three times a day just to ask, "Appa, did you take the tablets? Did you eat?"

One Tuesday afternoon he wakes from his nap later than usual, unsure whether he has already taken his afternoon tablet. He can't read the strip. He gets up to find out, the corridor is darker than he expected, and he stops halfway, holding the wall, waiting for someone to come home.

None of this was unpredictable. The house could have known he usually takes the tablet at 2 and hadn't yet. It could have known he was getting up, that the corridor was dark, that he was moving more slowly than usual. A device could have said, in his own language, before he had to wonder: "Appa, it's 3 o'clock — your afternoon tablet is the one in the blue box. I've switched on the corridor light."

This is not a problem of missing technology. Sensors are cheap, natural voices exist, and small boards can learn patterns. What is missing is **a device in the home that learns how a person lives, sees what they will need next, and helps — out loud and in the physical world — before they have to ask anyone.**

Shankar is not the exception, he is the rule. India has 14.9 crore people aged 60 and above (UNFPA India Ageing Report 2023), and older people are only one group. A person with low vision finding their way around a new office. A wheelchair user whose kitchen was designed for someone standing. Someone with hearing loss who misses the pressure-cooker whistle and the doorbell. A caregiver who can't be in the room all the time. Each of them depends on someone else for things they could do themselves, if the technology around them paid attention.

This challenge asks you to build that technology.

Three Failures. One Challenge.

Today's assistive technology fails people like Shankar in three ways at once:

THEY WAIT TO BE ASKED	Voice assistants and panic buttons only work when the person asks or presses. They never notice that the tablet wasn't taken, that the person is getting up in the dark, or that the usual evening routine hasn't started. The burden of remembering and asking stays with the person.
THEY LIVE ON A SCREEN	Most help arrives as a phone app: small text, small buttons, English, passwords and notifications. That is exactly what is hardest for someone with low vision, shaky hands or limited reading, so the help goes unused.
THEY CAN'T TOUCH THE WORLD	Even smart assistants mostly talk. They can't see that the corridor is dark, feel that the kettle is on, or switch on a light, buzz a caregiver or point the way. Help that stays in a speaker or a screen can't change what happens in the room.

Your Challenge

Build any assistive hardware device that uses voice (ElevenLabs) to predict what a person needs and help them before they ask, giving them more independence.

You choose the person and the freedom. Your device can be a wearable, a pendant, something on a walker or wheelchair, a box on the kitchen shelf, a pill organiser, a doorway unit or a bedside companion. It senses the person and their surroundings with real hardware, learns their routine, predicts what they will need, and acts or speaks before being asked.

A device that anticipates, not one that answers

Your device must work as a companion that pays attention over time, not a chatbot with a speaker. It senses, learns patterns, notices when something is about to be needed or about to go wrong, and speaks first.

It is a hardware device. A phone app, a web agent or a chatbot in a box does not meet this challenge. A phone may be one part of your system, but the sensing, prediction and help must come from the device.

Independence is the measure. At the end of your demo, the person you built for should be able to do something on their own that they used to need help with.

The Rules

- **It is a hardware product.** It senses the real world with real sensors of your choice. A phone or app can be part of it, but not the whole thing.
- **It speaks first.** At least once in the demo, your device acts or speaks before being asked.
- **Voice by ElevenLabs, in any language,** through any speaker, phone or laptop. Use whatever language your user actually speaks.
- **It does something real.** A physical action (a light, a buzzer, a display, a caregiver alert, a low-voltage device) or useful step-by-step guidance. Talking alone is not enough.
- **Build for someone real.** A specific person you have spoken to, not a generic user.

Building Blocks

Every anticipating assistive device needs these four pieces. They are building blocks, not a prescribed product: combine them in whatever form fits your person.

01 Sensing the Person and the Place

Use real sensors to capture what the person is doing and what is around them: a pill box opening, a door, movement in a room, someone getting out of bed, light level, a kettle or stove heating, sound, distance to an obstacle. Sample reliably and log the data — your prediction is only as good as what the device has seen.

Required: Real hardware sensing on your device. At least one physical sensor drives the prediction; a phone or web input alone does not count.

02 Anticipation

Learn the person's normal routine and predict the next need or the next problem: "The afternoon tablet is usually taken by 2:15 and the box hasn't opened", "He is getting up and the corridor is dark", "The cooker has whistled twice — the third is usually the one she turns it off on." This is where the hackathon's theme lives.

Required: A real prediction, as defined in The Prediction Requirement below.

03 The Voice

Design a voice the person actually wants to hear: warm, clear, the right pace, in their language. ElevenLabs supports text-to-speech, voice design and conversational agents. Think about how the device opens a conversation it started itself — politely, briefly, without startling anyone.

Required: Spoken output by ElevenLabs (credits are provided to every team), in any language.

04 Help in the Real World

Close the loop. Switch on a light, sound a buzzer, show a large-print display, alert a caregiver, guide someone step by step to the right tablet or the right door. Keep the person in control: anything involving money, health or safety needs a clear confirmation.

Required: A physical action on your hardware or useful spoken guidance that the person can act on, triggered by the prediction.

Stretch Goals

- **Graceful uncertainty** — the device says how sure it is and asks when it isn't.
- **Privacy by design** — learning happens on the device; no cameras unless the person truly wants one; only what is needed leaves the home.
- **Caregiver loop** — when a pattern breaks in a worrying way, the device knows whom to alert, and when, without nagging.

Directions You Could Explore

The problem is open. Pick a person and a freedom:

Spark people	Spark freedoms
Grandparent	Getting around alone
Wheelchair user	Handling money
New mother	Staying healthy
Person with low vision or hearing loss	Running the home
Farmer / shopkeeper	Staying connected
Caregiver	Anything else you find in your user interview

- **Wearables** — a pendant or wristband that notices a slowing walk or an unusual stillness and checks in by voice.
- **Medication** — a pill organiser that learns when each dose is usually taken, speaks up when one is about to be missed, and lights the right compartment.
- **Kitchen** — a stove-side unit that predicts the boil-over or the third cooker whistle and speaks before it happens, for someone who can't hear it or can't get there fast.
- **Indoor navigation** — a cane clip or doorway beacon that guides a person with low vision to the room they usually go to at this time.
- **Mobility aids** — a walker or wheelchair add-on that warns of an obstacle or a long, tiring route before the person sets off.
- **Home devices** — a bedside companion that learns night-time routines and lights the path before the person gets up.
- **Caregiver alerts** — a device that tells a caregiver when the routine is drifting, before it becomes an emergency, and tells the person what it has done.
- **Anything else** — if it helps someone do something on their own, it belongs here.

The Prediction Requirement

This is a Predictive Systems hackathon. A device that only answers is a search box with a voice.

Your device must make a **real prediction**. That means it forecasts something that has not happened yet, in one of these forms:

- **Time-to-event** — "about 40 seconds until this happens", updated as the situation changes.
- **An early-rising risk score** — a likelihood that climbs as precursors build up, well before the event itself.
- **A learned pattern** — the device learns what normal looks like for this person, place or machine, and flags the drift away from it before it becomes a problem.

Threshold alarms do not count. "Beep when the distance drops below 50 cm" or "buzz when the temperature crosses 40°C" is a reaction, not a prediction. A real forecast says what is coming and how soon, before any threshold is crossed.

Examples of what a real prediction looks like for this challenge:

- "Appa, your afternoon tablet is usually done by 2:15. It's in the blue box — I've lit it up for you." — triggered by a learned pattern from a pill-box sensor, not a fixed alarm.
- A pendant that notices the morning walk getting shorter and slower over several days and suggests a check-in call with the daughter.
- A bedside device that learns when the person usually gets up at night and switches on the path light just before they do.

Required: Show the device's forecast (what it expects, when, and how sure it is), show it speaking or acting first because of that forecast.

Build for Someone Real

Pick your own problem, but build it for a real person, not a generic "user". A grandparent in your family, a wheelchair user in your neighbourhood, a shopkeeper near campus, a farmer you can call — someone whose day you can actually describe.

Required: Talk to at least one real user before or during the hackathon (in person or on a call). Bring to judging: **one direct quote** from them, **one insight** you did not have before, and **one design change** that conversation caused.

The Demo Requirement

Hacktivate is judged on what works in front of the judges. There are no slides. Your demo is the device, your KiCad design and the person you built it for.

- **Live, no slides.** You present the working hardware on the table. A phone or laptop may show your sensor readings, forecast plot or serial monitor, but not a pitch deck.
- **KiCad circuit.** Show the schematic (and PCB layout, if you got that far) for the circuit you actually built. It should match the hardware on the table.
- **Working hardware.** A simulation, video or Wokwi model does not count as the demo.
- **Show it working.** Run a live demo of your product doing its job: sensing, making its prediction and acting on it.
- **It speaks first.** At least once, the assistant acts or speaks before anyone asks it to.

Required: All of the above, in a demo of about five minutes, followed by questions.

Resources

Voice and AI

- **ElevenLabs** — text-to-speech, voice design and conversational agents, with multilingual voices. Credits for every team. (elevenlabs.io/docs)

- **Speech-to-text** — any engine you like; test with real accents and background noise, not just your own voice in a quiet room.

Hardware: What You Get and What You Bring

- **Any hardware is allowed.** Any board, sensor, actuator, speaker, phone or laptop you can run safely; the form and the parts are your choice.
- **Provided by organisers:** ElevenLabs credits for every team. The PCB Cupid kit is reserved for Challenge 01 teams, so Challenge 02 teams bring their own hardware.
- **Bring your own:** whatever your idea needs. Most teams will want some way to sense the person or the place, some way to speak (any speaker, phone or laptop is fine), and power.
- **Ideas, not requirements** — teams often start with simple routine signals (a door or box opening, presence in a room, light level, a button the person already understands), but use whatever fits your person.

What Makes a Strong Submission

Strong Submissions...	Weak Submissions...
— Speak first, for a reason. The device starts the conversation because its sensors and forecast say something is coming — and can explain why.	— Build a chatbot in a box: a speaker that only responds when spoken to.
— Sense the real world. A pill box, a door, a stove, a footstep — real hardware feeds the prediction.	— Build a phone app or web agent and call it assistive hardware.
— Help physically. A light turns on, the right compartment glows, a caregiver gets a clear alert, the person is guided step by step.	— Just answer questions, or describe the help instead of giving it.
— Learn the routine. The device builds its expectations from what it senses and updates them when the routine changes.	— Hard-code a timetable of reminders and call it learning.
— Speak the person's language — literally, and in tone, pace and politeness. Bring their quote, the insight and the change it caused.	— Use a default English voice and design for "the elderly" or "the disabled" as a category, not a person.

Evaluation Criteria

Every challenge is judged out of 20 marks. Creativity breaks ties.

Criterion	Marks	Weight	What judges look for
KiCad design	5	25%	A clean schematic for the hardware you built (mic, speaker, sensors, controller, power) that matches the device on the table.
Hardware concept and working (incl. prediction quality)	7	35%	Does real hardware sense the person, does it make a real prediction, speak first via ElevenLabs and act or guide? Does it learn and adapt to the routine?
User understanding	3	15%	A real person, a real quote, a genuine insight and a design change that came from them.
Live demo	5	25%	Live, no slides; clear in about five minutes; the assistant speaks first and works in front of the judges.

Ground Rules

- **Keep people in control.** Anything involving money, health or safety needs a clear confirmation from the person.
- **Assistive and early-warning aids only.** No medical diagnosis claims. "Your posture is drifting" is fine; "you have a spinal condition" is not.

Why This Matters

Dependence is rarely about ability. Shankar can manage his tablets, his meals and his day. What he can't do easily is read small print, walk a dark corridor quickly or remember whether he already did something an hour ago. Each small thing he has to wait for someone else to do costs a little independence, and a little dignity.

Voice is the most natural interface people have. It works without reading, without small buttons and without English. Put it in a device that senses the home, learns a person's life and acts before they have to ask, and technology stops being one more thing to depend on others for. It becomes the thing that lets people depend on themselves. As India's population ages and more people spend their days alone at home, that may be one of the most useful things a builder can make.

Appendix — Contact & Additional Resources

Challenge contact: Ganesh S, Chair, IEEE SB, Ramaiah University of Applied Sciences · voxmastery@ambugo.tech · +91 79753 24769

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- ElevenLabs documentation: elevenlabs.io/docs
- Data: UNFPA India Ageing Report 2023 (india.unfpa.org).
- KiCad documentation: docs.kicad.org